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14. SAFETY INTERLOCKS

I-1, SHUT DOWN MAIN COMPRESSOR:

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock effect will shut down the main compressor 1540-GB-001.

CONFIGURATION:

Initiator(s):

Low-Low Waste Heat Boiler Steam Drum Level 1540-LSLL-7440 (2oo3: 1540-LSLL-7440A, 1540-LSLL-7440B, 1540-LSLL-7441)

OR

Common High-High Vibration / Thrust on Main Compressor (1540-VAHH-1060)

OR

Common High-High Bearing (Compressor, Gear Box, Motor) Temperature (1540-TAHH-1020)

OR

Low-Low Lube Oil Pressure of Main Compressor Lube Oil System (1540-PALL-1107)

OR

High-High Lube Oil Temperature (1540-TAHH-1096)

OR

DCS Initiated Shutdown (1540-HS-1071)

OR

Low-Low Acid Flow over Final Absorbing Tower unless bypassed at startup (1520-FSLL-6670)

OR

Low-Low Acid Flow over Interpass Absorbing Tower (1520-FSLL-6770)

OR

CONTROL ROOM Shutdown (1540-ESD-1094)

OR

Low-Low Drying Tower Discharge Pressure (1520-PSLL-5804).

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1540-PR-PID-0000-EXP-5040
1520-PR-PID-0000-EXP-5058
1520-PR-PID-0000-EXP-5066

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1520-PR-PID-0000-EXP-5067
1540-PR-PID-0000-EXP-5074
1520-PR-PID-0000-EXP-5114

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I-4, CLOSE MAIN COMPRESSOR DISCHARGE VALVE (1540-HV-4080):

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock will close main compressor discharge valve 1540-HV-4080.

CONFIGURATION:

Initiator(s):

Low-Low Main Compressor Discharge Pressure (1540-PSLL-4002)
AND
Shutdown of Main Compressor (1540-XS-1077).

Permissive (s):

None.

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1540-PR-PID-0000-EXP-5040

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I-7, STOP SULFUR FEED PUMPS AND CLOSE SULFUR FEED VALVE:

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock will stop sulfur feed pumps (1510-PP-004/005) and close sulfur feed valve (1540-FCV-2602).

CONFIGURATION:

Initiator(s):

High-High Sulfur Burner Outlet Temperature 1540-TSHH-4220 (2oo3: 1540-TSHH-4220A, 1540-TSHH-4220B, 1540-TSHH-4220C)

OR

Low-Low Waste Heat Boiler Steam Drum Level 1540-LSLL-7440 (2oo3: 1540-LSLL-7440A, 1540-LSLL-7440B, 1540-LSLL-7441)

OR

Shutdown of Main Compressor (1540-XS-1077)

OR

Low-Low Main Compressor Discharge Pressure (1540-PSLL-4002) and Low-Low Differential Pressure across Sulfur Burner and Waste Heat Boiler (1540-PDSLL-4200)

OR

Low-Low Flow to Tail Gas Scrubber Reverse Jet (1530-FSLL-0003) and Low-Low Pressure to Tail Gas Scrubber Reverse Jet (1540-PSLL-0004)

OR

All Tail Gas Scrubber Circulation Pumps Not Running (1530-PP-001/002)

OR

High-High Combination Pump Level (1540-LSHH-6040).

Permissive (s):

NONE

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1510-PR-PID-0000-EXP-5010
1520-PR-PID-0000-EXP-5060
1530-PR-PID-0000-EXP-5100
1530-PR-PID-0000-EXP-5102
1540-PR-PID-0000-EXP-5026
1540-PR-PID-0000-EXP-5040

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1540-PR-PID-0000-EXP-5042
1540-PR-PID-0000-EXP-5074

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I-8, SHUT DOWN STARTUP BURNER:

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock will shut down the Startup Burner via Startup Burner Management System 1540-PK-001.

CONFIGURATION:

Initiator(s):

Low-Low Differential Pressure across Sulfur Burner (1540-PDSLL-4200),
AND
Waste Heat Boiler (1540-PDSLL-4200) For >5 Seconds,
OR
High-High Sulfur Burner Outlet Temperature 1540-TSHH-4220 (2oo3: 1540-TSHH-4220A, 1540-TSHH-4220B, 1540-TSHH-4220C),
OR
Operator initiated shutdown, Remote ESD (1540-XE-2993),
OR
Shutdown of Main Compressor (1540-XS-1077)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1540-PR-PID-0000-EXP-5029
 1520-PR-PID-0000-EXP-5040

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I-9, ISOLATE DRYING/INTERPASS TOWER DILUTION WATER:

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock will close Dilution Valve (1520-ACV-6060), open Bleed Valve (1520-XV-6081) and closes Block Valves (1520-XV-6071 & 1520-XV-6080).

CONFIGURATION:

Initiator(s):

Shutdown of Main Compressor (1540-XS-1077)

OR

Low-Low acid concentration strength on Drying & Interpass Towers Analyzer (1520-ASLL-6060),

OR

Shutdown of Drying & Interpass Towers Acid Pump (1520-PP-001).

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1520-PR-PID-0000-EXP-5060
 1540-PR-PID-0000-EXP-5040

NOTE: When Interlock (I-9) is active, enable the Flow Alarm 1520-FSH-6070.

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I-10, ISOLATE FINAL TOWER DILUTION WATER:

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock will close Dilution Valve (1520-ACV-6063), open Bleed Valve (1520-XV-6082) and closes Block Valves (1520-XV-6072 & 1520-XV-6083).

CONFIGURATION:

Initiator(s):

Shutdown of Main Compressor (1540-XS-1077)

OR

Low-Low acid concentration strength on Final Towers Analyzer (1520-ASLL-6063)

OR

Shutdown of Final Towers Acid Pump (1520-PP-002).

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1520-PR-PID-0000-EXP-5060
 1540-PR-PID-0000-EXP-5040

NOTE: When Interlock (I-10) is active, enable the Flow Alarm 1520-FSH-6074.

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I-13, CLOSE COMBINATION PUMP TANK LEVEL CONTROL VALVE (1520-LCV-6040A):

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock will close Dilution Valve (1520-ACV-6040A).

CONFIGURATION:

Initiator(s):

Low-Low Combination Pump Tank Level (1520-LSLL-6040)

OR

Drying Tower/Interpass Tower Acid Pump not running (1520-PP-001)

OR

High-High Dilution Pump Tank Level (1520-LSHH-6240)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1520-PR-PID-0000-EXP-5060
 1520-PR-PID-0000-EXP-5062
 1520-PR-PID-0000-EXP-5067

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I-14, ISOLATE ACID TO PRODUCT AREA:

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock closes Dilution Pump Tank Level Control Valve (1520-LCV-6240), shuts down Product Acid Pump (1520-PP-005), shuts down Startup Acid Unloading Pump (1570-PP-007) and closes Startup Acid Unloading Pump Inlet Valve (1570-HV-9011).

CONFIGURATION:

Initiator(s):

Low-Low Dilution Pump Tank Level (1520-LSLL-6240)

OR

High-High Product Acid Storage Tank 1570-TK-0001 Level (1570-LSHH-8601)

OR

High-High Product Acid Storage Tank 1570-TK-0002 level (1570-LSHH-8604)

OR

High-High Startup Acid Storage Tank (1570-TK-0003) Level (1570-LSHH-9009)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1520-PR-PID-0000-EXP-5062
 1520-PR-PID-0000-EXP-5064
 1570-PR-PID-0000-EXP-5186
 1570-PR-PID-0000-EXP-5190

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I-15, CLOSE DILUTION PUMP TANK LEVEL CONTROL VALVE (1520-LCV-6240):

DESCRIPTION:

Upon activation of the below initiator, this interlock closes Dilution Pump Tank Level Control Valve (1520-LCV-6240).

CONFIGURATION:

Initiator(s):

Product Acid Pump Shutdown (1520-PP-005)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1520-PR-PID-0000-EXP-5062
 1520-PR-PID-0000-EXP-5064

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I-16, ISOLATE PRODUCT ACID DILUTION WATER:

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock closes Inlet Water Valve (1520-ACV-6260), opens Bleed Valve (1520-XV-6081), and closes Block Valves (1520-XV-6280 & 1520-XV-6082).

CONFIGURATION:

Initiator(s):

Low-Low acid concentration strength on Product Acid Analyzer (1520-ASLL-6260)

OR

Shutdown of Drying Tower/Interpass Tower Acid Pump (1520-PP-001)

OR

Shutdown of Product Acid Pump (1520-PP-005)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1520-PR-PID-0000-EXP-5060
 1520-PR-PID-0000-EXP-5062

NOTE: When Interlock 16 (I-16) is active, enable the Flow Alarm 1520-FSH-6270.

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I-21, SHUT DOWN HP BOILER FEEDWATER PUMPS (1540-PP-017, 1540-PP-018):

DESCRIPTION:

Upon activation of the below initiator, this interlock shuts down HP Boiler Feedwater Pumps 1540-PP-017 and 1540-PP-018.

CONFIGURATION:

Initiator(s):

Low-Low Deaerator (1540-PV-002) Level (1540-LSLL-6842)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1540-PR-PID-0000-EXP-5068
 1540-PR-PID-0000-EXP-5070

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I-23, AUTOSTART STANDBY HP BOILER FEEDWATER PUMPS (1540-PP-017, 1540-PP-018):

DESCRIPTION:

Upon activation of the below initiator, this interlock shuts down HP Boiler Feedwater Pumps 1540-PP-017 and 1540-PP-018.

CONFIGURATION:

Initiator(s):

Low-Low HP Boiler Feedwater Discharge Pressure (1540-PSLL-7000)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1540-PR-PID-0000-EXP-5070

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I-60, SHUT DOWN TAIL GAS SCRUBBER CIRCULATION PUMPS (1530-PP-001 & 1530-PP-002) AND CLOSE SCRUBBER EFFLUENT VALVE (1530-DCV-0006):

DESCRIPTION:

Upon activation of the below initiator, this interlock shuts down Tail Gas Scrubber Pumps 1530-PP-001 and 1530-PP-002 and closes Scrubber Effluent Valve 1530-DCV-0006.

CONFIGURATION:

Initiator(s):

Low-Low Tail Gas Scrubber Level (2oo2: 1530-LSLL-0013 and 1530-LSLL-0014)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1530-PR-PID-0000-EXP-5100
 1530-PR-PID-0000-EXP-5102

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I-61, CLOSE TAIL GAS SCRUBBER MAKE-UP WATER VALVE (1530-LCV-0013):

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock closes Tail Gas Scrubber Make-Up Water Valve 1530-LCV-0013.

CONFIGURATION:

Initiator(s):

High-High Tail Gas Scrubber Level (1530-LSHH-0013)

OR

High-High Tail Gas Scrubber Level (1530-LSHH-0014)

OR

Tail Gas Scrubber Circulation Pumps (1530-PP-001 & 1530-PP-002) not running.

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1530-PR-PID-0000-EXP-5100
 1530-PR-PID-0000-EXP-5102

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I-62, CLOSE CAUSTIC SUPPLY CONTROL VALVE (1530-ACV-0002):

DESCRIPTION:

Upon activation of the below initiator, this interlock closes Tail Gas Scrubber \ Caustic Supply Control Valve 1530-ACV-0002.

CONFIGURATION:

Initiator(s):

Tail Gas Scrubber Circulation Pumps (1530-PP-001 & 1530-PP-002) not Running.

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1530-PR-PID-0000-EXP-5100
 1530-PR-PID-0000-EXP-5102

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I-65, SHUT DOWN OXIDATION AIR BLOWERS (1530-GB-001 AND 1530-GB-002):

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock shuts down the Oxidation Blowers 1530-GB-001 and 1530-GB-002.

CONFIGURATION:

Initiator(s):

High-High Oxidation Air Temperature to Tail Gas Scrubber (1530-TSHH-0654)

OR

High-High Oxidation Air Discharge Pressure to Tail Gas Scrubber (1530-PSHH-0652)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1530-PR-PID-0000-EXP-5106

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I-83, OPEN TREATED WATER BYPASS VALVE 1520-XV-6742, CLOSE WATER BLOCK VALVES (1520-XV-6741 & 1520-XV-6738), AND SWITCH 1520-TIC-6731 TO MANUAL WITH OUTPUT 0 TO CLOSE ACID VALVE 1520-TCV-6731:

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock opens water bypass valve (1520-XV-6742), closes water block Valves (1520-XV-6741 and 1520-XV-6738) and switch the controller 1520-TIC-6731 to manual close acid valve 1520-TCV-6731.

CONFIGURATION:

Initiator(s):

High-High BFW Preheater Treated Water Outlet Temperature (1520-TSHH-6731) to Deaerator,
OR

High-High BFW Preheater Treated Water Outlet Conductivity (1520-ASHH-6745) to Deaerator,
OR

High-High BFW Preheater Treated Water Outlet Conductivity (1520-ASHH-6746) to Deaerator

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1520-PR-PID-0000-EXP-5067

Resume Normal Operation:

Once both the High-High Conductivity (1520-ASHH-6745 & 1520-ASHH-6746) and High-High Temperature (1520-TSHH-6731) conditions have cleared, press Resume Button (or perform manually) to:

- Close valve 1520-XV-6742,
- Open valves 1520-XV-6741 and 1520-XV-6738,
- Return controller 1520-TIC-6731 to Automatic mode.

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I-100, LIGHT ALARM BEACON 1510-XA-1118A, SET 1510-LIC-1007 TO MANUAL AND OUTPUT TO 0 TO CLOSE SULFUR FEED VALVE 1510-LCV-1007:

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock will turn on the visual alarm/beacon (1510-XA-1118A) to alert the operators of the high level of unloading pit and stop filling the pit.

Level controller 1510-LIC-1007 will be set to manual to force to close the sulfur feed valve 1510-LCV-1007.

CONFIGURATION:

Initiator(s):

High-High Sulfur Pit Level (1510-LSHH-1007A)

OR

High-High Sulfur Pit Level (1510-LSHH-1007B)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1510-PR-PID-0000-EXP-5010

1510-PR-PID-0000-EXP-5011

Resume Normal Operation:

Once the sulfur pit level returns to normal (below HH setpoint), the DCS will automatically restore control:

- Return controller 1510-LIC-1007 to Automatic mode

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I-101, STOP LIQUID SULFUR UNLOADING PUMPS 1510-PP-001 AND 1510-PP-002:

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock will stop the liquid sulfur unloading pumps (1510-PP-001 and 1510-PP-002).

CONFIGURATION:

Initiator(s):

High-High Sulfur Storage Tank Level (1510-LSHH-1201)

OR

High-High Sulfur Storage Tank Level (1510-LSHH-1202)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1510-PR-PID-0000-EXP-5010
 1510-PR-PID-0000-EXP-5012

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I-110, STOP FIN FAN COOLING WATER PUMPS 1520-PP-008, 1520-PP-009, 1520-PP-010:

DESCRIPTION:

Upon activation of the below initiator, this interlock will stop the Fin Fan Cooling Pumps (1520-PP-008, 1520-PP-009 and 1520-PP-010).

CONFIGURATION:

Initiator(s):

Low-Low Fin Fan Cooling Water Expansion Tank Level (1520-LSLL-1401)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1520-PR-PID-0000-EXP-5114

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I-111, STOP ACID PLANT COOLING TOWER PUMPS 1550-PP-001, 1550-PP-002:

DESCRIPTION:

Upon activation of the below initiator, this interlock will stop Acid Plant Cooling Tower Pumps 1550-PP-001 and 1550-PP-002.

CONFIGURATION:

Initiator(s):

Low-Low Cooling Tower Basin Level (1550-LSLL-2215)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1550-PR-PID-0000-EXP-5122

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I-112, STOP TREATED WATER EXPORT PUMPS 1560-PP-006, 1560-PP-007, 1560-PP-010 AND 1560-PP-011:

DESCRIPTION:

Upon activation of the below initiator, this interlock will stop Treated Water Export Pumps 1560-PP-006, 1560-PP-007, 1560-PP-010 and 1560-PP-011.

CONFIGURATION:

Initiator(s):

Low-Low Treated Water Storage Level (1560-LSLL-3401)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5134
 1560-PR-PID-0000-EXP-5138

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I-115, START BYPASS CONDENSATE PUMP 1560-PP-019 OR 1560-PP-020:

DESCRIPTION:

Upon activation of the below initiator, this interlock will start Bypass Condensate Pumps 1560-PP-019 or 1560-PP-020 (standby).

CONFIGURATION:

Initiator(s):

Low-Low Booster Pump Discharge Pressure (1560-PSLL-7524)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5176

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I-121, OPEN OFF-SPEC CONDENSATE RETURN VALVE 1560-XV-7513, CLOSE CONDENSATE RETURN TO BFW PREHEATER VALVE 1560-XV-7514:

DESCRIPTION:

Upon activation of the below initiator, this interlock will open the Off Spec Condensate Return On/Off Valve 1560-XV-7513 and closes the Condensate return to the BFW Preheater Valve 1560-XV-7514.

CONFIGURATION:

Initiator(s):

High-High Condensate Conductivity Alarm (1560-ASHH-7512) with one (1) minute delay

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5176

Resume Normal Operation:

Once the condensate conductivity (1560-AI-7512) returns to normal (below 1560-ASHH-7512 setpoint), the DCS will automatically restore control:

- Open on/off valve 1560-XV-7514
- Close on/off valve 1560-XV-7513

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I-123, SWITCH 1560-TCV-3603 TO MANUAL MODE WITH OUTPUT TO ZERO AND SWITCH 1560-TCV-3604 TO MANUAL MODE WITH 100% OPEN TO BYPASS THE CONDENSATE TO THE WATER TREATMENT ARE SUMP ON:

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock will close TCV-3102A and opens TCV-3102B to bypass the condensate to the water treatment sump.

Delays the opening of TCV-3102B to give enough time for the liquid level in the tank to fall below an acceptable level and the temperature to drop down as well.

CONFIGURATION:

Initiator(s):

High-High Liquid Level (1560-LSHH-2901) in Recovery Tank (1560-TK-005)

OR

High-High Temperature on Raw Water header (TIT-3102)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5129
 1560-PR-PID-0000-EXP-5136

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I-124, STOP WATER RECOVERY PUMPS (1560-PP-021 AND 1560-PP-022):

DESCRIPTION:

Upon activation of the below initiator, this interlock will stop Water Recovery Pumps 1560-PP-021 and 1560-PP-022.

CONFIGURATION:

Initiator(s):

Low-Low Recovery Tank Water Level (1560-LSLL-2901)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5129

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I-126, STOP ACC CONDENSATE PUMPS (1560-PP-015 AND 1560-PP-016), STOP BYPASS CONDENSATE PUMPS (1560-PP-019 AND 1560-PP-020):

DESCRIPTION:

Upon activation of the below initiator, this interlock will stop Condensate Pumps (1560-PP-015 and 1560-PP-016) and Stops Bypass Condensate Pumps (1560-PP-019 and 1560-PP-020).

CONFIGURATION:

Initiator(s):

Low-Low Condensate Tank Water Level (1560-LSLL-9320)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5175
 1560-PR-PID-0000-EXP-5176

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I-140, CLOSE SULFUR VAPOR SCRUBBER BLOWDOWN VALVE (1510-HV-1512) ON:

DESCRIPTION:

Upon activation of the below initiator, this interlock will close the Sulfur Vapor Scrubber Valve (1510-XV-1512).

CONFIGURATION:

Initiator(s):

Low-Low Sulfur Vapor Scrubber Liquid Level (1510-LSLL-1524)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1510-PR-PID-0000-EXP-5015

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I-141, CLOSE MAKE UP WATER VALVE (1510-LCV-1524) ON:

DESCRIPTION:

Upon activation of the below initiator, this interlock will close the Make up Water Valve (1510-LCV-1524).

CONFIGURATION:

Initiator(s):

High-High Sulfur Vapor Scrubber Liquid Level (1510-LSHH-1524)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1510-PR-PID-0000-EXP-5015

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I-150, STOP 98.5% ACID RECIRCULATION PUMPS (1570-PP-005 AND 1570-PP-006):

DESCRIPTION:

Upon activation of the below initiator, this interlock will stop the Acid Recirculation Pumps 1570-PP-005 and 1570-PP-006.

CONFIGURATION:

Initiator(s):

Low-Low Start-up Acid Storage Tank Level (1570-LSLL-9009)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1570-PR-PID-0000-EXP-5188
 1570-PR-PID-0000-EXP-5190

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I-151, CLOSE ACID LOADING VALVE (1570-HV-9202) ON:

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock will close the Acid Loading Valve (1570-HV-9202) and shuts down Acid Loading pumps (1570-PP-003 and 1570-PP-004).

CONFIGURATION:

Initiator(s):

Totalizer quantity is greater than Set Point (1570-FQI-9201),
OR
Acid loading arm high Level limit switch triggered (1570-LAHH-9203),
OR
Arm 1570-ZM-001 not in position (1570-ZA-9204),
OR
High-High acid loading area sump level (1570-LSHH-9215),
OR
Truck not Grounded (1570-XI-9230),
OR
Flow Control Valve actuator fail closed (1570-FCV-9201),
OR
Local Emergency Shutdown Push Button Triggered (1570-HS-9202B)
OR
Local Emergency Shutdown Push Button Triggered (1570-HS-9212B)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1570-PR-PID-0000-EXP-5186
1570-PR-PID-0000-EXP-5192

	Interlock Narrative Sulfuric Acid Plant		Revision	A
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I-152, CLOSE ACID LOADING VALVE (1570-HV-9212) ON:

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock will close the Acid Loading Valve (1570-HV-9212) and shut down the Acid Loading Pumps (1570-PP-003 and 1570-PP-004).

CONFIGURATION:

Initiator(s):

Totalizer > Set Point (1570-FQI-9211),
 OR
 Acid loading arm high Level limit switch triggered (1570-LAHH-9213,
 OR
 Arm 1570-ZM-001 not in position (1570-ZA-9214),
 OR
 High-High acid loading area sump level (1570-LSHH-9215),
 OR
 Truck not Grounded (1570-XI-9231),
 OR
 Flow Control Valve actuator failure (1570-FCV-9211),
 OR
 Local Emergency Shutdown Push Button Triggered (1570-HS-9202B),
 OR
 Local Emergency Shutdown Push Button Triggered (1570-HS-9212B)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1570-PR-PID-0000-EXP-5186
 1570-PR-PID-0000-EXP-5192

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I-153A, STOP ACID HEATER (1570-ZE-003) ON:

DESCRIPTION:

Upon activation of the below initiator, this interlock will stop the Acid Heater 1570-ZE-003.

CONFIGURATION:

Initiator(s):

Low-Low Acid Flow (1570-FSLL-8816)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1570-PR-PID-0000-EXP-5188

	Interlock Narrative Sulfuric Acid Plant		Revision	A
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I-154A, STOP ACID HEATER (1570-ZE-004) ON:

DESCRIPTION:

Upon activation of the below initiator, this interlock will stop the Acid Heater 1570-ZE-004.

CONFIGURATION:

Initiator(s):

Low-Low Acid Flow (1570-FSLL-8815)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1570-PR-PID-0000-EXP-5188

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I-155, CLOSE LEVEL CONTROL VALVE (1520-LCV-6240), TRUCK UNLOADING ON/OFF VALVE (1570-HV-9011), START-UP ACID STORAGE INLET ON/OFF VALVE (1570-HV-9010), AND STOP START-UP ACID UNLOADING PUMP (1570-PP-007) ON:

DESCRIPTION:

Upon activation of the below initiator, this interlock will close 1520-LCV-6240, 1570-HV-9011, 1570-HV-9010 and stop start the Acid Unloading Pump 1570-PP-007.

CONFIGURATION:

Initiator(s):

High-High Start-Up Acid Storage Tank (1570-TK-003) Level (1570-LSHH-9009)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1570-PR-PID-0000-EXP-5190
 1570-PR-PID-0000-EXP-5192

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I-156, PRODUCT ACID TRANSFER PUMP (1570-PP-001 AND 1570-PP-002):

DESCRIPTION:

Upon activation of the below initiator, this interlock will start the Product Acid Transfer Pumps 1570-PP-001 and 1570-PP-002.

CONFIGURATION:

Initiator(s):

N/A

Permissive (s):

Lithium Processing Plant must give permission to start the pumps.

(NEED TO HAVE THE LITHIUM PROCESS PLANT READY SIGNAL TAG NUMBER)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1570-PR-PID-0000-EXP-5186

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I-160, STOP CAUSTIC UNLOADING PUMP (1510-PP-003) ON:

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock will stop the Caustic Unloading Pump 1510-PP-003.

CONFIGURATION:

Initiator(s):

High-High Caustic Tank (1510-TK-003) Level (1510-LSHH-9516)

OR

Truck Not Grounded (1570-XI-9540)

OR

Emergency Shutdown Push Button Triggered (1510-HS-9512)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1570-PR-PID-0000-EXP-5182

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I-161, STOP CAUSTIC FEED PUMPS (1510-PP-007 AND 1510-PP-008) ON:

DESCRIPTION:

Upon activation of the below initiator, this interlock will stop the Caustic Feed Pump 1510-PP-007 and 1510-PP-008 (Standby).

CONFIGURATION:

Initiator(s):

Low-Low Caustic Tank (1510-TK-003) Level (1510-LSLL-9516)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1570-PR-PID-0000-EXP-5182

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I-162, START STANDBY CAUSTIC FEED PUMP (1510-PP-007 OR 1510-PP-008) ON:

DESCRIPTION:

Upon activation of the below initiator, this interlock will start the Caustic Feed Pump 1510-PP-007 or 1510-PP-008, whichever is on standby.

CONFIGURATION:

Initiator(s):

Low Caustic Supply Header Pressure (1510-PSL-9529)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1570-PR-PID-0000-EXP-5195

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I-167, STOP SA1 EFFLUENT FEED PUMPS 1520-PP-023 AND 1520-PP-024 AND SA1 EFFLUENT STORAGE TANK AGITATOR 1520-AG-003 ON:

DESCRIPTION:

Upon activation of the below initiator, this interlock will stop SA1 Effluent Feed Pumps 1520-PP-023 and 1520-PP-024 SA1 Effluent Storage Tank Agitator 1520-AG-003.

CONFIGURATION:

Initiator(s):

Low-Low Effluent Storage Tank Level (1520-LSLL-4705)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1520-PR-PID-0000-EXP-5147

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I-169, STOP TRANSLOADING EFFLUENT UNLOADING PUMP 1520-PP-021 AND SET 1520-XV-4711 TO MANIAL MODE WITH ZERO OUTPUT ON:

DESCRIPTION:

Upon activation of any of the below initiators, this interlock will stop Transloading Effluent Unloading Pump 1520-PP-021 and set 1520-XV-4711 to manual mode.

CONFIGURATION:

Initiator(s):

High-High Effluent Storage Tank Liquid Level (1520-LSHH-4705),
OR
Emergency Shutdown Push Button Triggered (1520-HS-9010B)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1520-PR-PID-0000-EXP-5147

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I-170, CLOSE CONTROL VALVE 1520-LCV-4705 ON:

DESCRIPTION:

Upon activation of the below initiator, this interlock will close the control valve 1520-LCV-4705.

CONFIGURATION:

Initiator(s):

Low-Low Effluent Storage Tank pH level (1520-ASHH-4711),

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1520-PR-PID-0000-EXP-5147

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I-301, TRIP STEAM TURBO GENERATOR AND SET BYPASS DESUPERHEATER VALVE 1560-PCV-6672 TO MANUAL MODE WITH ZERO OUTPUT ON:

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock will trip the Steam Turbo Generator and Set the Bypass Desuperheater Valve 1560-PCV-6672 to manual close with zero output.

CONFIGURATION:

Initiator(s):

High-High Temperature of Exhaust Stem (2003, 1560-TSHH-9312),
OR
High Temperature of Exhaust Steam (1560-HS-9312) and timer (Note 1)
OR
High-High Pressure of exhaust steam (2003, 1560-PSHH-9303),
OR
High-High Condensate Tank (1560-TK-004) Pressure (1560-PSHH-9323),
OR
High-High Condensate Tank (1560-TK-004) liquid level (2003, 1560-LSHH-9320),
OR
High-High Drain Pot Level (1560-LSHH-9407),
OR
DCS Emergency Trip (1560-XS-6210)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5174
 1560-PR-PID-0000-EXP-5160
 1560-PR-PID-0000-EXP-5166
 1560-PR-PID-0000-EXP-5175

NOTE:

1. Please refer to the figure below as recommended in 1500-IN-VEN-3410-EXP-1201_Functional Description for Air Cooler Condenser Section 2.3 for timber setting vs temperature.

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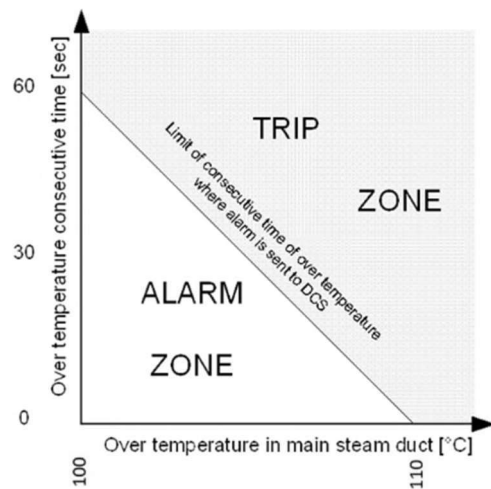


Figure 1 – ACC Trip/Alarm limit

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I-303, INITIATE AIR COOLED CONDENSER (ACC) UNIT FREEZE PROTECTION SEQUENCE:

- I. EVERY 15 MINUTES THE FAN ASSEMBLY OF THE MIX MODULE (1560-AN-001/002/003), IF IT IS RUNNING, WILL BE ALTERNATIVELY SLOWED DOWN BY 20 HZ OR STOPPED IF THE FREQUENCY IS BELOW 15 HZ (MIN VALUE TO START THE FAN) FOR A 10-MINUTE PERIOD TO KEEP THE TUBES WARM.
- II. IF ANY OF THE CONDENSATE TEMPERATURE READING (1560-TE-9326 OR 1560-TE-9328) IS LOWER THAN 25°C (77°F), THEN AN ALARM IS TRIGGERED AT DCS AND THE TURBINE BACKPRESSURE SET-POINT SHALL BE INCREASED 30 MBAR (ADJUSTABLE) EVERY 30 MINUTES UNTIL ALL CONDENSATE TEMPERATURES HAVE RISEN OVER 30°C (86°F). WHEN ALL CONDENSATE TEMPERATURES (1560-TE-9326 OR 1560-TE-9328) ARE OVER 30°C (86°F), THE BP SET-POINT SHALL BE DECREASED 30 MBAR (ADJUSTABLE) EVERY 60 MINUTES UNTIL THE ORIGINAL SET POINT IS REACHED.
- III. TURNING ON HEAT TRACING (1560-HS-9331) FOR INSTRUMENTS, LINES AND VALVES.

CONFIGURATION:

Initiator(s):

Ambient Air Temperature (1560-TE-9300) is lower than - 3°C (26.6°F), for a continuous period of 5 minutes.

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5174

NOTE:

1. Resume Action: The ACC unit freeze protection function will be deactivated when the ambient temperature is higher than 3°C (37.4°F), for a continuous period of 5 minutes.
2. During heating periods, the fan control step shall not be changed.

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I-304, AIR COOLED CONDENSER (ACC) UNIT FREEZE PROTECTION TRIP SEQUENCE:

- I. TRIP STEAM TURBO GENERATOR AND SET BYPASS DESUPERHEATER VALVE 1560-PCV-6672 TO MANUAL MODE WITH ZERO OUTPUT.
- II. SHUTDOWN FAN ASSEMBLY (1560-AN-001, 1560-AN-002 AND 1560-AN-003)
- III. OPEN SKY VENT 1560-PCV-6018

CONFIGURATION:

Initiator(s):

Ambient Air Temperature (1560-TSL-9300) is lower than - 3°C (26.6°F), for a continuous period of 5 minutes,
AND
Low Air-Cooled Condenser Unit flowrate (1560- FSL-6004),
AND
Low Ambient Temperature (1560-TSL-9300)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5166
 1560-PR-PID-0000-EXP-5174
 1560-PR-PID-0000-EXP-5160

NOTE:

Please refer to the figure below as recommended in 1500-IN-VEN-3410-EXP-1201_Functional Description for Air Cooler Condenser Section 2.3 for minimum flow/temperature limit.

Temperature (°F)	Minimum flow (lb/h)
32	38581
23	52911
14	67242
5	84879
0.9	93697

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I-310, TRIP AIR COOLED CONDENSER (ACC) FAN MOTORS ON:

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock will trip the associated Air-Cooled Condenser (ACC) Fan Motor.

CONFIGURATION:

Initiator(s):

1560-AN001

HIGH-HIGH MOTOR VIBRATION (1560-VSHH-9382),

OR

HIGH-HIGH GEAR BOX TEMPERATURE (1596-TSHH-9397 OR 1560-TSHH-9402)

OR

HIGH-HIGH MOTER TEMPERATURE (1560-TSHH-9357, 1560-TSHH-9362, 1560-TSHH-9367, 1560-TSHH-9372, 1560-TSHH-9377)

1560-AN002

HIGH-HIGH MOTOR VIBRATION (1560-VSHH-9383),

OR

HIGH-HIGH GEAR BOX TEMPERATURE (1596-TSHH-9398, 1560-TSHH-9403)

OR

HIGH-HIGH MOTER TEMPERATURE (1560-TSHH-9358, 1560-TSHH-9363, 1560-TSHH-9368, 1560-TSHH-9373, 1560-TSHH-9378)

1560-AN003

HIGH-HIGH MOTOR VIBRATION (1560-VSHH-9384),

OR

HIGH-HIGH GEAR BOX TEMPERATURE (1596-TSHH-9399, 1560-TSHH-9404)

OR

HIGH-HIGH MOTER TEMPERATURE (1560-TSHH-9359, 1560-TSHH-9364, 1560-TSHH-9369, 1560-TSHH-9374, 1560-TSHH-9379)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5174

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I-312, TRIP AIR COOLED CONDENSER (ACC) HOGGING SKID LIQUID RING PUMP (1560-AP-001)
ON:

DESCRIPTION:

Upon activation of any of the below initiators, this interlock will trip the ACC pump 1560-AP-001.

CONFIGURATION:

Initiator(s):

High-High ACC Hogging Skid Separation Tank Liquid Level (1560-LSHH-9333),

OR

Low-Low ACC Hogging Skid Separator Tank (1560-BB-002) Liquid Level (1560-LSLL-9334) for more than 180 seconds.

AND

Closes V1560-9329 upon High-High Separator Tank Level (1560-LSHH-9333)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5177

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I-313, OPEN AIR-COOLED CONDENSER (ACC) HOGGING SKID SEPARATOR FILLING VALVE (V1560-9329) ON:

DESCRIPTION:

Upon activation of the below initiator, this interlock will open the ACC Hogging Skid Separation Valve (V1560-9329).

CONFIGURATION:

Initiator(s):

Low-Low ACC Hogging Skid Separator Tank (1560-BB-002) Liquid Level (1560-LSLL-9334) for more than 120 seconds.

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5177

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I-314, TRIP AIR COOLED CONDENSER (ACC) SKID EJECTOR (1550-BN011 OR 1550-BN012) ON:

DESCRIPTION:

Upon activation of any of the below initiators, this interlock will trip the ACC Skid Ejector 1550-BN011 or 1550-BN012.

CONFIGURATION:

Initiator(s):

High-High Motive Steam Temperature (1560-TSHH-9353),
OR
High-High Motive Steam Pressure (1560-PSHH-9352),
OR
Low-Low Gland Steam Condenser Flow (1560-FSLL-7511).

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5176
1560-PR-PID-0000-EXP-5177

NOTE:

1560-FSLL-7511 will be triggered when flowrate < 40 t/h (88,185 lbs/h) for 30s (Steam Turbine operation), or < 120 t/h (264,544 lbs/h) for 30s (Bypass operation)

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I-315, FORCE TO CLOSE ALL AIR SUCTION VALVES (V1560-9340) ON:

DESCRIPTION:

Upon activation of the below initiator, this interlock will close the Air Suction Valve (V1560-9340).

CONFIGURATION:

Initiator(s):

Low-Low Motive Steam Pressure (1560-PSLL-9352)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5177

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I-316, FORCE TO CLOSE ALL MOTIVE STEAM VALVES (V1560-9363, V1560-9366) ON:

DESCRIPTION:

Upon activation of the below initiator, this interlock will close the Motive Steam Valves V1560-9340 and V1560-9366).

CONFIGURATION:

Initiator(s):

High-High Motive Steam Pressure (1560-PSHH-9352)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5177

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I-319, TRIP AIR COOLED CONDENSER (ACC) AIR REMOVAL SYSTEM ON:

DESCRIPTION:

Upon activation of any of the below initiators, this interlock will trip the ACC Air Removal System.

CONFIGURATION:

Initiator(s):

Holding skid Motive Steam Inlet Valves (V1560-9363 or V1560-9366) Fault,
OR
Holding skid air suction valve (V1560-9340) Fault,
OR
Hogging skid air inlet valve (V1560-9324) Fault,
OR
Hogging skid water refill valve (V1560-9329) Fault.

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5177

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I-322, TRIP STEAM TURBO GENERATOR AND CHANGE BYPASS DESUPERHEATER VALVE (1560-PCV-6672) TO AUTO MODE ON:

DESCRIPTION:

Upon activation of any of the below initiators, this interlock will trip the Steam Turbo Generator and change the Bypass Desuperheater Valve (1560-PCV-6672) mode to Auto.

CONFIGURATION:

Initiator(s):

Main Compressor Trip (1540-XS-1077)

OR

High-High Waste Heat Boiler Level (2oo3, 1540-LSHH-7440)

OR

DCS Emergency Trip (1560-XS-6210)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1540-PR-PID-0000-EXP-5040
 1540-PR-PID-0000-EXP-5074
 1560-PR-PID-0000-EXP-5160
 1560-PR-PID-0000-EXP-5166

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I-324, STEAM TURBO GENERATOR (STG) BYPASS PROTECTION:

DESCRIPTION:

This interlock will set the Bypass Valve (1560-PCV-6672) into manual mode with zero (0) output. It will only be active when the Steam Turbo Generator Bypass protection signal (1560-XS-6202) is available.

CONFIGURATION:

Initiator(s):

High-High STG last stage pressure (1560-PIA-6317)
OR
High-High STG last stage temperature (1560-TIA-6232)
AND
Bypass Protection Signal (1560-XS-6202)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5160

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15. PERMISSIVES

I-2, MAIN COMPRESSOR START PERMISSIVE

DESCRIPTION:

The interlock permits the Main Compressor to start only when all the following conditions are satisfied.

CONFIGURATION:

Permissive(s):

Main compressor discharge valve 1540-XV-4080 fully open (1540-ZI-4080).

AND

Main compressor inlet guide vanes closed (1540-ZAC-4030).

AND

field-mounted shutdown hand switch disengaged (1540-ESD-1094).

AND

Acid flow over drying tower not low (1540-FSLL-5870).

AND

All hydrogen vents confirmed closed (1540-ZAC-5805, 1540-ZAC-6630, 1540-ZAC-8421).

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1540-PR-PID-0000-EXP-5040
 1520-PR-PID-0000-EXP-5058
 1520-PR-PID-0000-EXP-5066
 1520-PR-PID-0000-EXP-5084

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I-80, HOT HEAT EXCHANGER BYPASS DAMPERS (1540-TCV-5224A & 1540-TCV-5224B) HEAT UP MODE PERMISSIVE (1540-TIC-5224):

DESCRIPTION:

The interlock permits Heat-up mode (Start-up Bypass mode) from DCS (1540-HS-5224) when all the following conditions are satisfied.

CONFIGURATION:

Permissive(s):

Main Compressor (1540-GB-001) Not Running (1540-XS-1077)
AND
Sulfur Feed Pump (1510-PP-004) Not Running
AND
Standby Sulfur Feed Pump (1510-PP-005) Not Running

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1510-PR-PID-0000-EXP-5010
 1540-PR-PID-0000-EXP-5040
 1540-PR-PID-0000-EXP-5052

Note: Allows manual closure of 1540-TCV-5524A and 1540-TCV-5524B when start up duct is in place.

The valves shall be programmed in the DCS to prevent closing both during normal operation and in manual mode.

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I-81, OPEN HYDROGEN VENTS (1520-XV-5805, 1520-XV-6630, 1520-XV-8423) PERMISSIVE:

DESCRIPTION:

To remove the Hydrogen built up inside the Tower's dome, the On/Off Valves will be in open position during shutdown mode. The interlock permits opening hydrogen vents (1520-XV-5805, 1520-XV-6630, 1520-XV-8423) when all the following conditions are satisfied.

CONFIGURATION:

Permissive(s):

Sulfur Feed Pumps (1510-PP-004 & 1510-PP-005) Not Running 1510-HS-1004A & 1510-HS-1004B),

AND

Main Compressor Speed (1540-XI-1079) less than 200 RPM.

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1510-PR-PID-0000-EXP-5010
 1520-PR-PID-0000-EXP-5058
 1520-PR-PID-0000-EXP-5066
 1520-PR-PID-0000-EXP-5084
 1540-PR-PID-0000-EXP-5040

NOTE: The valves must be in closed position during normal operation which will function as permissive to initiate interlock I-164. This will allow the Sulfur Feed Pumps (1510-PP-004 and 1510-PP-005) start-up.

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I-156, PRODUCT ACID TRANSFER PUMPS 1570-PP-001/002 START PERMISSIVE:

DESCRIPTION:

The interlock permits the Product Acid Transfer Pumps (1570-PP-001/002) to start only when all the following conditions are satisfied.

CONFIGURATION:

Permissive(s):

Process Plant Ready Indication

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1570-PR-PID-0000-EXP-5186

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I-164, SULFUR FEED PUMPS 1510-PP-004 AND 1510-PP-005 START PERMISSIVE ON:

DESCRIPTION:

Below initiators must be satisfied as permissive to start the Sulfur Feed Pumps 1510-PP-004 and 1510-PP-005.

CONFIGURATION:

Permissive(s):

Drying Tower Vent Valve Closed (1520-ZAC-5805)
AND
Interpass Tower Vent Valve Closed (1520-ZAC-8423)
AND
Final Tower Vent Valve Closed (1520-ZAC-6630)
AND
Tail Gas Scrubbed pH Normal (1530-AIC-0002)
AND
{(Oxidation Blowers Running (1530-GB-001)
OR
Oxidation Blowers Running (1530-GB-002))
AND
Waste Heat Boiler Not Low or High (1540-HS-7440)
AND
Sulfur Burner Outlet Temperature Not Low (1540-TSL-4220)
AND
Main Compressor Running (1540-XS-1077)
AND
Fin Fan Cooling Water Flow Not Low, 1520-FSL-1413
AND
Cooling Tower Water Flow Not Low, 1550-FSL-2218
AND
Treated Water Pump (1560-PP-006 OR 1560-PP-007) Running
AND
Hot Heat Exchanger Bypass Dampers Not Both Closed (1530-TCV-5224A AND 1530-TCV-5224B)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1520-PR-PID-0000-EXP-5058
 1520-PR-PID-0000-EXP-5084
 1520-PR-PID-0000-EXP-5066

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1530-PR-PID-0000-EXP-5100
1530-PR-PID-0000-EXP-5106
1540-PR-PID-0000-EXP-5074
1550-PR-PID-0000-EXP-5122
1520-PR-PID-0000-EXP-5114
1560-PR-PID-0000-EXP-5134
1540-PR-PID-0000-EXP-5052

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I-171, TRANSLOADING EFFLUENT UNLOADING PUMP 1520-PP-021 START PERMISSIVE ON:

DESCRIPTION:

The interlock permits the Transloading Effluent Unloading Pumps (1520-PP-021) to start (1520-HS-4702) only when the following condition are satisfied.

CONFIGURATION:

Permissive(s):

Effluent Storage Tank Liquid Level Below High Setpoint (1520-LSH-4705).

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1520-PR-PID-0000-EXP-5147

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I-200, 20% CAUSTIC SOLUTION EXPORT TO LITHIUM PROCESSING PLANT (ON/OFF VALVE 1510-XV-9530) PERMISSIVE ON:

DESCRIPTION:

The interlock permits 20% caustic solution export to lithium processing plant (ON/OFF Valve 1510-XV-9530) only when all the following conditions are satisfied.

CONFIGURATION:

Initiator(s):

Permissive from Lithium Processing Plant Ready Indication (XXXX-XX-XXXX).

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1510-PR-PID-0000-EXP-5195

	Interlock Narrative Sulfuric Acid Plant		Revision	A
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I-201, TREATED WATER EXPORT PUMPS 1560-PP-006 AND 1560-PP-007 (STANDBY) START PERMISSIVE ON:

DESCRIPTION:

The interlock permits treated water export pumps 1560-PP-006/007 to start only when all the following conditions are satisfied.

CONFIGURATION:

Permissive(s):

Permissive from Lithium Processing Plant Ready Indication (XXXX-XX-XXXX).
NEED TO HAVE THE LITHIUM PROCESS PLANT READY SIGNAL TAG NUMBER)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5134

	Interlock Narrative Sulfuric Acid Plant		Revision	A
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I-202, 150 PSIG STEAM EXPORT STATION (1560-PCV-6011) LITHIUM PROCESSING PLANT START PERMISSIVE ON:

DESCRIPTION:

The interlock permits 150PSIG Steam Station (1560-PCV-6011) to lithium process plant to start only when all the following conditions are satisfied.

CONFIGURATION:

Permissive(s):

Permissive from Lithium Processing Plant Ready Indication (XXXX-XX-XXXX).
NEED TO HAVE THE LITHIUM PROCESS PLANT READY SIGNAL TAG NUMBER)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5165

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I-302, PERMISSIVE TO OPEN BYPASS DESUPERHEATER VALVE 1560-PCV-6672:

DESCRIPTION:

The interlock permits opening bypass desuperheater valve 1560-PCV-6672 only when the following condition is satisfied.

CONFIGURATION:

Permissive(s):

PRESSURE OF EXHAUST STEAM LINE > L (1560-PSL-9304)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5174
 1560-PR-PID-0000-EXP-5165

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I-307, AIR COOLED CONDENSER (ACC) UNIT FAN START UP PERMISSIVES

DESCRIPTION:

The interlock permits air cooled condenser (ACC) unit fan start only when all the following condition is satisfied.

CONFIGURATION:

Permissive(s):

ACC condenser air take-off Temperature (1560-TIA-9330) > Ambient temperature (1560-TIA-9300) +50°F

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5174

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I-308, PERMISSIVE FOR AIR COOLED CONDENSER (ACC) START-UP SEQUENCE:

DESCRIPTION:

The interlock permits ACC Start-up Sequence only when all the following conditions are satisfied.

CONFIGURATION:

Permissive(s):

High Condensate Tank Level (1560-LSLL-9320)
AND
Fan Assembly in Auto Mode (1560-AN-001/1560-AN-002/1560-AN-003)
AND
{(Condensate Extraction Pump (1560-PP-015) Running
OR
Condensate Extraction Pump (1560-PP-016) Running)}
AND
Gland Steam Condenser Flow NOT Low-Low (1560-FSLL-7511).
AND
Air Removal System is in Auto Mode (1560-XA-XXXX)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5174
 1560-PR-PID-0000-EXP-5175
 1560-PR-PID-0000-EXP-5176

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I-317, PERMISSIVE FOR OPENING ACC HOLDING SKID AIR SUCTION VALVE(V1560-9340) ON:

DESCRIPTION:

The interlock permits opening ACC holding skid air suction valve (V1560-9340) only when the following condition is satisfied.

CONFIGURATION:

Permissive(s):

Main Holding Skid Motive Steam Valve (V1560-9363) is open.

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5177

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I-320, PERMISSIVE FOR OPENING EXHAUST STEAM LINE VACUUM BREAKER (V1560-9308) ON:

DESCRIPTION:

The interlock permits opening exhaust steam line vacuum breaker (V1560-9308) only when the following condition is satisfied.

CONFIGURATION:

Permissive(s):

Steam Turbo Generator Speed (1560-XS-6201) < 300 rpm

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5174

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I-323, PERMISSIVE FOR PUT FAN ASSEMBLY FUNCTION GROUP IN AUTO ON:

DESCRIPTION:

The interlock permits putting fan assembly function group in AUTO mode only when all the following conditions are satisfied.

CONFIGURATION:

Permissive(s):

ALL motors (1560-AN001/1560-AN002/156-AN003) available in Auto Mode.

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5174

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16. ON-OFF CONTROL INTERLOCKS

I-63A, OPEN RAW WATER TO OXIDATION BLOWERS AIR SPARGER ON/OFF VALVE (1530-XV-0656):

I-63B, CLOSE RAW WATER TO OXIDATION BLOWERS AIR SPARGER ON/OFF VALVE (1530-XV-0656):

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock will open and close the On/Off Valve 1530-XV-0656.

CONFIGURATION:

Initiator(s):

Oxidation Blowers (1530-GB-001 or 1530-GB-002) is Running'

THUS:

On/Off Valve 1530-XV-0656 opens.

OR

Oxidation Blowers (1530-GB-001 or 1530-GB-002) is Not Running

THUS:

On/Off Valve 1530-XV-0656 closes.

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1530-PR-PID-0000-EXP-5106

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I-103, START/STOP LIQUID SULFUR UNLOADING PUMPS 1510-PP-001 AND 1510-PP-002:

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock will start/stop the liquid sulfur unloading pumps (1510-PP-001 and 1510-PP-002).

CONFIGURATION:

Initiator(s):

- START SULFUR UNLOADING PUMPS 1510-PP-001/1510-PP-002 ON:
 - A. HIGH LEVEL OF SULFUR PIT (1510-LSH-1007A OR 1510-LSH-1007B)
- STOP SULFUR UNLOADING PUMPS 1510-PP-001/1510-PP-002 ON:
 - B. LOW SULFUR UNLOADING PIT LEVEL (1510-LSL-1007A OR 1510-LSL-1007B)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1510-PR-PID-0000-EXP-5010

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I-117 START / STOP WATER TREATMENT AREA SUMP PUMPS 1560-PP-012 OR 1560-PP-013:

DESCRIPTION:

Upon activation of the below initiator, this interlock will stop the Area Sump Pump 1560-PP-012 or 1560-PP-013.

CONFIGURATION:

Initiator(s):

- ON HIGH-HIGH SUMP LEVEL (1560-LAHH-3601):
START BOTH WATER TREATMENT AREA SUMP PUMPS (1560-PP-012 & 1560-PP-013)
- ON HIGH SUMP LEVEL (1560-LSH-3601):
START ONLY ONE WATER TREATMENT AREA SUMP PUMPS (1560-PP-012 OR 1560-PP-013)
- ON LOW SUMP LEVEL (1560-LSL-3601):
START ONLY ONE WATER TREATMENT AREA SUMP PUMPS (1560-PP-012 OR 1560-PP-013)
- ON LOW-LOW SUMP LEVEL (1560-LALL-3601):
AN ALARM WILL BE INITIATED IN DCS

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5136

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I-119, START BOILER BLOWDOWN SUMP PUMP 1540-PP-001:

DESCRIPTION:

Upon activation of the below initiator, this interlock will start the Boiler Blowdown Sump Pump 1540-PP-001.

CONFIGURATION:

Initiator(s):

High Boiler Blowdown Sump Level (1540-LSH-7655)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1540-PR-PID-0000-EXP-5076

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I-120, STOP BOILER BLOWDOWN SUMP PUMP 1540-PP-001:

DESCRIPTION:

Upon activation of the below initiator, this interlock will stop the Boiler Blowdown Sump Pump 1540-PP-001.

CONFIGURATION:

Initiator(s):

Low Boiler Blowdown Sump Level (1540-LSL-7655)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1540-PR-PID-0000-EXP-5076

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I-122, OPEN/CLOSE DEAERATOR OVERFLOW VALVE 1540-XV-6843:

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock will open the Deaerator Overflow On/Off Valve 1540-XV-6843.

CONFIGURATION:

Initiator(s):

- OPEN DEAERATOR OVERFLOW VALVE 1540-LV-6843 ON:
 - A. HIGH-HIGH DEAERATOR LEVEL (**1540-LSHH-6841 & 1540-LSH-6840 SHOULD THIS BE 1002 OR 2002?**)
- CLOSE DEAERATOR OVERFLOW VALVE LV-6843
 - B. NORMAL DEAERATOR LEVEL (1540-LSH-6841)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5176

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I-137, START AND STOP ACID PLANT COOLING TOWER FAN (1550-CT-001) ON:

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock will start and stop Cooling Tower Fan (1550-CT-001).

CONFIGURATION:

Initiator(s):

Stops the Fan on Cooling Water Header Temperature Low (1550-TSL-2430),
OR
Starts the Fan on Cooling Water Header Temperature High (1550-TSH-2430).

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1550-PR-PID-0000-EXP-5124

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I-142, SULFUR VAPOR SCRUBBER (1510-PK-007) FLUSH SEQUENCE STARTS ON:

- Close Make-up Water Valve (1510-LCV-1524). Status is confirmed with 1510-XIC-1524,
- Open Scrubber Drain Valve (1510-XEV-1512) for XX seconds,
- Open Water Chase Valve (1510-XEV-1526),
- Close Water Chase Valve (1510-XEV-1526) after Scrubber Drain Valve (1510-XEV-1512) closed. Status is confirmed with 1510-XIC-1512,
- Dis-engage Sulfur Vapor Scrubber Flush Hand Switch (1510-HS-1512).

DESCRIPTION:

Upon activation of the below initiator, above flushing operations will begin.

CONFIGURATION:

Initiator(s):

Sulfur Vapor Scrubber Flush Hand Switch Engaged (1510-HS-1512)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1510-PR-PID-0000-EXP-5015

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I-153B, START/STOP ACID HEATER (1570-ZE-003) ON:

DESCRIPTION:

Upon activation of the below initiator, this interlock will start the Acid Heater 1570-ZE-003.

CONFIGURATION:

Initiator(s):

- STOP ACID HEATER (1570-ZE-003) ON:
 - A. HIGH ACID HEATER 1570-ZE-003 TEMPERATURE (1570-TSH-8806A)
- START ACID HEATER (1570-ZE-003)
 - B. LOW ACID HEATER 1570- ZE-003 TEMPERATURE (1570-TSL-8806A)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1570-PR-PID-0000-EXP-5188

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I-154B, START/STOP ACID HEATER (1570-ZE-003) ON:

DESCRIPTION:

Upon activation of the below initiator, this interlock will start the Acid Heater 1570-ZE-004.

CONFIGURATION:

Initiator(s):

- STOP ACID HEATER (1570-ZE-004) ON:
 - A. HIGH ACID HEATER 1570- ZE-004 TEMPERATURE (1570-TSH-8806B)
- START ACID HEATER (1570- ZE-004)
 - B. LOW ACID HEATER 1570-ZE-004 TEMPERATURE (1570-TSL-8806B)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1570-PR-PID-0000-EXP-5188

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I-163, STARTUP BURNER SYSTEM PURGE FORCE ON:

DESCRIPTION:

This interlock will monitor the Startup Burner Purge Force is active and will be monitored by DCS signal 1540-XA-2990.

CONFIGURATION:

Initiator(s):

- I. SET MAIN COMPRESSOR CAPACITY CONTROL (1540-HIC-4030) TO 40% OUTPUT, ENABLES 1540-FS-4070 AND SEND SIGNAL LOGIC "1" TO BMS ONCE FLOWRATE IS REACHED ON:
 - A. PURGE FORCE SIGNAL IS RECEIVED FROM BMS (1540-XI-2990)
- II. SET MAIN COMPRESSOR CAPACITY CONTROL (1540-HIC-4030) TO 0% OUTPUT AND DISABLES FS- 4070 ON:
 - B. PURGE FORCE SIGNAL IS REVOKED FROM BMS (1540-XI-2990)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1540-PR-PID-0000-EXP-5029

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I-318, STARTUP BURNER SYSTEM PURGE FORCE ON:

DESCRIPTION:

This interlock will monitor the Startup Burner Purge Force is active and will be monitored by DCS signal 1540-XA-2990.

CONFIGURATION:

Initiator(s):

- OPEN ACC HOLDING UNIT FIRST STAGE EJECTOR (1560-BN-011/012) ON:
A. STEAM/AIR PRESSURE HIGH (1560-PAH-9336)
- CLOSE ACC HOLDING UNIT FIRST STAGE EJECTOR (1560-BN-011/012) ON:
B. STEAM/AIR PRESSURE LOW (1560-PAL-9336)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5177

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I-321, CLOSE EXHAUST STEAM LINE VACUUM BREAKER (V1560-9308) ON:

DESCRIPTION:

This interlock will close Exhaust Steam Line Vacuum Breaker (V1560-9308) ON.

CONFIGURATION:

Initiator(s):

STEAM TURBO GENERATOR SPEED (1560-XS-6201) IS LESS THAN 300 RPM

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5174

DOCUMENT END